MILITARY SPECIFICATION

AMPLIFIER SYSTEM, AUDIO, LOW NOISE, HIGH INPUT IMPEDANCE

1. SCOPE

1.1 This specification covers requirements for low noise, high input impedance audio amplifier systems for use with remote capacitive transducers.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-915 - Cable, Electrical, Special Purpose, General Specification for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirement for.
MIL-C-5015 - Connectors, Electric, "AN" Type.
MIL-S-7742 - Screw Threads, Standard, Optimum Selected Series: General Specification for.
MIL-P-15024 - Plates, Identification--Information and Marking for Identification of Electrical and Electronic and Mechanical Equipment.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts, Packaging and Packing of.
MIL-E-21981 - Electronics Type Designation, Identification Plates and Markings; Requirements for.
MIL-C-22857 - Connectors, Electrical, Rectangular, Crimp Type, Removable Contact for Rack and Panel and Other Applications: General Specification for.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-130 - Identification Markings of U.S. Military Property.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-189 - Racks, Electrical Equipment, 19-Inch and Associated Panels.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462 - Electromagnetic Interference Characteristics Measurement of.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNITED STATES OF AMERICA STANDARDS INSTITUTE

USAS S1.4-1961 - General Purpose Sound Level Meters.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to production, a sample shall be tested as specified in 4.2 (see 6.2).

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3.2 General requirements. The amplifier system shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

General requirements
Primary requirements
NAVSEC approval
Completeness of equipment
Omissions, mistakes and discrepancies
Changes and developments
Definitions
Materials, general
Materials, substitution of
Fungus-inert materials
Flammable materials
Arc-resistant materials
Toxic materials
Metals
Aluminum
Magnesium
Ferrous alloys
Non ferrous material (except aluminum)
Springs, material
Other metals
Plastics
Mechanical parts
Electrical insulating parts
Laminated thermosetting plastics
Molded thermosetting plastics
Thermoplastic materials
Painted plastics
Plastic identification plates
Dials and transparent or translucent parts
Ceramics
Impregnating, embedding, and encapsulating compounds
Selection of material
Glass
Painting
Cleaning
Finish coats
Protection against corrosion
Corrosion resisting materials
Corrosion resisting treatments
Selections of metals
Bolts, machine screws, studs and nuts
Parts, mechanical
Gaskets
Locking devices
Shaft locking devices
Lock washers
Washers
Bearings
Bearings, Ball
Bearings, sleeve-type
Parts, electrical
Use of nonstandard parts
Semi-conductor devices
Electron tubes
Electron tube or capacitor sockets
Capacitors
Variable resistors
Transformers and inductors
Relays
Servo motors
Electrical tapes
Batteries (see 3.4.1.8)
Switches
Toggle switches
Sensitive switches
Rotary switches
Power switches

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Push switches
Fuses
Fuseholders
Fuse clips
Printed wiring
Features, mechanical
Enclosures, general
Enclosure, accessibility
Enclosure, degree of (see 3.4.4.1, 3.4.4.2.1, 3.4.4.3.1, and 3.4.4.4.1)
Enclosure, mounting (see 3.4.4)
Universal mounting, enclosure
Stand mounting
Panel mounting
Stiffening grooves
Minimum sheet metal thickness
Through bolting
Ventilation
Size (see 3.4.4.2.3)
Threaded devices
Thread cutting screws
Flat head screws
Threads in aluminum (corrosion-resistant steel inserts permitted)
Thread projection
Threads in plastics
Rounded corners and edges
Internal subassembly protection
Structural welding
Temperature and humidity
Extended storage
Extreme temperature for exposed location
Shock and vibration (vital)
Shock and vibration mountings
Shock and vibration mountings
Features, electrical
Overload protection
Primary power supply circuits (switch and fusing required)
Power supply tolerances (see 3.4.1.8)
Safety (personnel hazard)
Shielding and radio frequency noise reduction (see 3.6)
Bonding, electrical
Ground potential and grounding
Soldering
Electrical parts mounting
Internal subassembly connection
Electrical connectors
Terminal boards
Terminal
Braided wire cable shielding
Mounting terminal boards
Terminal and terminal board marking
Wiring
Hook-up wire
Transformer and inductance wire
Leads
Magnet wire and insulation
Wiring arrangement
Sleeving insulation
Color coding of internal wiring
Electrical insulation (except class O will not be permitted)
Limitations on use of classes of insulation
Material standards of insulation
Insulating electrical windings
Varnish restrictions
Varnish treatment procedures
Dielectric strength and insulation resistance clearances
Miscellaneous requirements
Drawings, General
Drawings, preliminary
Drawings, working
Clarity and legibility of drawings
Design changes after drawing approval
Schematic diagrams
Wiring diagram

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Drawing list
Assembly drawings
Drawings, manufacturing
Bill of material
Interchangeability and standardization
Equipment standardization
Standard stock parts
Proprietary parts
Manuals
Drawings in manuals
Manuals for developmental contracts
Repair parts (electronic)
Special tools (not permitted)
Designations and marking (serial numbers required)
Designations of parts
Reports
Item names and nomenclature

3.3 General description. The amplifier system shall consist of the following subunits (see Figure 1):

(a) Preamplifier - Conditions capacitive type self-generative transducer signals for transmission over cable 500 feet minimum in length. The type of preamplifier shall be as specified (see 6.2).

(b) Signal Normalizer - Adjustable sensitivity and full scale range for a 1-volt full scale low impedance output. Contains the Signal Calibrator unit.

(c) Signal Calibrator - An oscillator signal source normally mounted in the Signal Normalizer chassis that may be removed for portable use. Provides battery power to the entire system when installed. Also provides stowage and power to the Vibration Calibrator and Microphone Calibrator units.

(d) Vibration Calibrator - An accelerometer vibrator for calibration, driven by the Signal Calibrator generating 1 g root-mean-square (r.m.s.) (9.81 meters per sec^2 or 386 inches per sec^2).

(e) Microphone Calibrator - Generates a precise sound pressure level and is driven by the Signal Calibrator.

3.3.1 Subunits (d), and (e) shall not be supplied unless specified (see 6.2), however the Signal Calibrator shall be built to provide for (d) and (e).

3.4 Operating characteristics

3.4.1 Amplifier system.

3.4.1.1 Range. The ratio of the Signal Normalizer output r.m.s. voltage to the preamplifier r.m.s. voltage or charge input shall be constant for any combinations of sensitivity and range control settings and for any preamplifier input signal within the following ranges to the specified tolerances:

(a) Frequency ranges - 5 Hertz (HZ) to 10 kiloHertz (kHz) ± 5 percent
2 Hz to 20 kHz ± 15 percent

(b) Amplitude range - 10 microvolts (uV) r.m.s. to 1 volt r.m.s. with voltage amplifier type preamplifier
5×10^{-3} r.m.s. to 500 picocoulombs r.m.s. with charge amplifier type preamplifier.

3.4.1.2 Residual noise. The amplifier system shall not generate noise having an r.m.s. level above 3 uV (1.5×10^{-3} r.m.s. picocoulombs) in any 1/3 octave band centered at any frequency from 7 Hz to 10 kHz.

3.4.1.3 Output noise level. Internal noise, when measured in octave bands shall be 40 decibels (dB) or more below any full scale output level.

3.4.1.4 Phase shift. Phase shift of any sinusoidal signal within the ranges specified in 3.4.1.1 shall be a constant function of frequency. The constant shall vary not more than plus or minus 10 percent.

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3.4.1.5 Transient recovery time. The system shall recover to the residual noise flow specified in 3.4.1.2 from a negative direct current (d.c.) voltage level equal to 10 times any full scale input to the preamplifier in less than 5 seconds and in less than 2 seconds from an equal positive input d.c. signal. This requirement shall be met for any sensitivity setting.

3.4.1.6 Signal calibration unit. A 1 volt r.m.s. ± 1 percent; 125, 250, 500, 1000, and 2000 Hz ± 1 percent oscillator shall be provided for output calibration. Selection of 0.1, 0.01, and 0.001 volts r.m.s. ± 10 percent shall also be provided.

3.4.1.7 Signal normalizer output impedance. The Signal Normalizer output impedance shall be less than 100 ohms in series with more than 100 microfarads.

3.4.1.8 Power supply modes. The following power supply modes shall be provided:

(a) Normal - 115 volts, 60 cycle in accordance with MIL-I-983.
(b) Battery - (1) Amplifier - 12-hour continuous operation within specified performance requirements.
(2) Vibration calibrator - 1/2-hour continuous operation with a 300 gram load at 1 g and 125 Hz.
(3) Microphone calibrator - 1/2-hour continuous operation.
(c) Battery charge - 14 hours to recharge from fully discharged conditions using the input power source specified in (a).

3.4.1.9 Meter. A Signal Normalizer output meter with movement characteristics in accordance with paragraph 4.4 of USAS S1.4-1961 shall be furnished. The meter detector shall have a full-wave rectifier. A simple peak detector shall also be provided.

3.4.1.10 Vibration calibrator. A 125 Hz ± 1 percent for 50 gram load and 125 Hz + 0, -2 percent for 300 gram load 1 g r.m.s. Vibration Calibrator shall be provided when specified (see 6.2). Capability to calibrate any pick-up weighing from 50 to 300 grams shall be provided. The unit driving power shall be supplied from the Signal Calibration unit.

3.4.1.11 Microphone calibrator. When specified (see 6.2), a microphone calibrator shall be provided that can be driven by the Signal calibrator unit to produce a sound pressure level of 114 ± 3 dB referred to 10 dynes per square centimeter at 125, 250, 500, and 1000 Hz, and down 12 dB at 2000 Hz.

3.4.1.12 Preamplifier gain. Unless otherwise specified (see 6.2), the preamplifier gain shall be 20 and 40 dB.

3.4.2 Controls.

3.4.2.1 Preamplifier gain selection. Unless otherwise specified (see 6.2), a two position switch shall be provided on the preamplifier for selection of 20 or 40 db gain.

3.4.2.2 Sensitivity adjustment. A Signal Normalizer continuous attenuator that is readable and repeatable to one part in 1000 shall provide 1 volt r.m.s. output from preamplifier input signals of 0.3×10^{-3}V, or less to 1500×10^{-3}V r.m.s., or more for a voltage-type preamplifier or 0.15, or less to 750 or more, picocoulombs' r.m.s. for a charge-type preamplifier.

3.4.2.3 Range control. A stepped Signal Normalizer attenuator to change output range or scale factor by plus 0, 10, 20 and 30 dB and by minus 10, 20, 30, 40, and 60 dB shall be provided.

3.4.2.4 Signal Normalizer meter. The Signal Normalizer output meter scale shall be at least 3 inches long and shall have a linear 0 to 10 scale and logarithmic 0 to 20 and 0 to 3 dB scales. The meter shall have a simple peak detector operated manually by a "hold down" button that resets or returns to normal operation when released.

3.4.2.5 Signal Calibration unit. The Calibration output source specified in 3.4.1.6 shall have an "ON-OFF" switch that interrupts all power and voltage sources to the oscillator. The calibration oscillator shall contain the battery power supply and shall be removable without tools from the Signal Normalizer's front panel (see 3.4.4.2.2). A battery check switch and meter to indicate calibration signal level as well as battery checks shall be provided. The meter shall also contain scale marking necessary for use with the Vibration Calibrator specified in 3.4.1.10.

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3.4.2.6 Vibration Calibrator. The Vibration Calibrator level adjustment shall be designed to operate by using the meter on the Signal Calibrator.

3.4.2.7 Power switch. An "EXT-OFF-BAT-CHG" Multi-position switch shall be provided on the Signal Normalizer. Power circuitry shall be in accordance with 3.4.1.8.

3.4.3 Connectors.

3.4.3.1 Preamplifier signal input. The signal input connector shall be coaxial with the shield connection electrically isolated. The fittings shall be #10-32NF-2A in accordance with MIL-S-7742 and an attached threaded metal protective cap shall be provided. (Input to take a Microdot 52-173 cable assembly or equivalent.) The input shield isolation to the preamplifier body shall be 50 megohms minimum and 100 picofarads maximum.

3.4.3.1.1 Calibration or filter section input. A parallel input connector with protective cap, identical to the connector specified in 3.4.3.1, shall be provided.

3.4.3.1.2 Signal output. An "AN" type, submersible Class E connector with protective cap, excluding quick disconnect types, in accordance with MIL-C-5015 shall be provided.

3.4.3.1.3 Signal output cable assembly. When specified (see 6.2), a 50-foot length of portable 3-shielded pair cable in accordance with MIL-C-915 with a mating connector for the preamplifier signal output on one end of the cable and the quick disconnecting type connector on the other end shall be provided. Unless otherwise specified (see 6.2), the latter connector shall be Amphenol 91-853, or equivalent. Protective metallic connector cover caps shall be attached to each end of the cable.

3.4.3.2 Signal Normalizer.

3.4.3.2.1 Signal input. One AN type, class E connector as specified in MIL-C-5015 (excluding quick disconnect types) shall be mounted on the rear panel and a parallel front panel connector that mates directly with the preamplifier output connector. Threaded metal protective caps shall be chained to the unit for each connector.

3.4.3.2.2 Power input. The power cord shall enter the rear of the Signal Normalizer. The length shall be 4 feet and shall conform to the power plugs and primary power supply circuits requirements of MIL-I-983.

3.4.3.2.3 Signal output. Unless otherwise specified (see 6.2), one signal output connector phone jack type BRS-3-12, or equivalent shall be provided on the front face and one parallel "AN" type shall be provided on the rear face.

3.4.3.2.4 Normalizer output to Analyzer cable assembly. When specified (see 6.2), a 2-foot cable assembly mating with the signal output connector (see 3.4.3.2.3) on one end and having a three-pin microphone type quick connector Amphenol 91-853, or equivalent, shall be provided on the other end.

3.4.3.2.5 Signal Calibrator unit. Front face connectors with attached protective metal caps in accordance with MIL-C-22857 or MIL-C-5015 shall be provided for connection between the Signal Calibration unit and the Signal Normalizer unit or the Vibration Calibration unit.

3.4.3.2.5.1 Vibration Calibrator. A front face oscillator input connector shall be provided on each Vibration Calibrator unit. The connector shall be identical to the type used on the Signal Normalizer unit for connection between the Signal Normalizer and the Signal Calibrator units so that the same cable assembly may be used for connection of the Signal Calibrator to the Vibration Calibrator units or the Signal Normalizer unit.

3.4.3.2.5.2 Signal Calibrator cable assembly. A 2-foot cable assembly for interconnection between the Signal Calibrator and Signal Normalizer units and between the Signal Calibrator and Vibration Calibrator units shall be furnished .

3.4.4 Physical characteristics.

3.4.4.1 Preamplifier enclosure. The preamplifier shall be magnetically shielded with 1/32-inch thick "CO-NETIC" alloy, or equivalent. The housing material may be corrosion-resistant steel or aluminum, but a low carbon annealed (1500°F for 20 minutes, uniform cool rate of 150°F per hour) steel is preferred for additional magnetic shielding. The preamplifier unit shall be watertight in accordance with MIL-STD-108. A removable shock absorbing insulating boot shall be required with each preamplifier. Ready removal and insertion of the preamplifier from the boot without disconnecting the preamplifier input

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cable, and without the use of tools shall be provided.

3.4.4.1.1 Preamplifier mounting. The preamplifier unit shall be designed such that it can be mounted "in line" with the insulating boot.

3.4.4.2 Signal Normalizer.

3.4.4.2.1 Enclosure. The Signal Normalizer unit shall be dripproof in accordance with MIL-STD-108.

3.4.4.2.2 Mounting. The Signal Normalizer unit shall be suitable for front face cantilever mounting using a rack designed in accordance with MIL-STD-189.

3.4.4.2.3 Dimensions. Unless otherwise specified (see 6.2), the dimensions for the Signal Normalizer unit shall be:

(a) Height - 6-1/2 inches maximum
(b) Depth - 13-1/4 inches maximum

3.4.4.2.4 Accessory stowage. The Signal Normalizer unit shall provide for stowage of the following:

(a) One pick-up assembly - The preamplifier and an accelerometer connected together by 3 feet of low noise coaxial cable. Provide a maximum of 20 inches ^{3}.
(b) Space for one accelerometer, two insulating mounting studs, and two permanent magnets. Provide a maximum of 8 inches ^{3}.
(c) One spare Preamplifier.

3.4.4.3 Signal Calibrator.

3.4.4.3.1 Enclosure. The Signal Calibrator unit shall be dripproof and dripproof protected in accordance with MIL-STD-108. The unit shall have carrying handles for portable use.

3.4.4.3.2 Mounting. The Signal Calibrator unit shall mount into the Signal Normalizer's front panel. The attachments shall be such that the unit may be readily removed for portable use. All controls shall be accessible when mounted in the Signal Normlizer unit.

3.4.4.3.3 Power supply. The Signal Calibrator unit shall contain the batteries to supply the entire amplifier system. The Signal Normalizer unit shall contain the battery charger.

3.4.4.3.4 Stowage. The Signal Calibrator unit shall be designed to stow one Vibration Calibrator unit and one Microphone Calibrator unit plus their interconnecting cables.

3.4.4.4 Vibration Calibrator.

3.4.4.4.1 Enclosure. The Vibration Calibrator unit shall be dripproof and dripproof protected in accordance with MIL-STD-108 and shall have carrying handles for portable use.

3.4.4.4.2 Mounting. The Vibration Calibrator unit shall be locked in place inside the Signal Calibrator chassis when not in use. The unit shall be stowable and removable without the use of tools.

3.4.4.4.3 Power supply. Power shall be supplied to the Vibration Calibrator unit from the Signal Calibrator unit.

3.5 Environmental. The operating characteristics specified in 3.4 shall be attained during any of the environmental conditions specified in 4.4.1 for the accelerated life test excluding vibration, under any operating condition and control setting.

3.6 Electromagnetic interference emission and susceptibility. The amplifier system shall withstand the electromagnetic interference tests for type IIC equipment as specified in MIL-STD-461.

3.7 Identification plates. Identification plates shall be type F, size 9 or larger, in accordance with MIL-P-15024 and MIL-E-21981. The plates shall be stowed in a closed slot accessible from the Signal Normalizer's front face. Space and attachment for one additional

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card for Navy use shall be provided. The "drawout" cards shall be fastened so they cannot be detached from the unit unless the housing is removed. A spare set of identification cards shall be furnished and shall be stowed in each Signal Normalizer unit on a side or bottom housing panel. Special characteristics as specified in MIL-STD-130 shall include the following information on the plates:

(a) System ranges for less than _____% change in output/input ratio at any control setting (all volts and currents are RMS)

Frequency - _____Hz to _____KHz
Input Amplitude - _____microvolts to _____volts

(b) System noise floor _____ microvolts over above band measured without further filtering.

(c) Normalizer

	Capacitance	Resistance
Max. input load		
Min. input load		
Max. output load		
Min. output load		
Input		
Output		
Output full scale voltage	_____ volts	
APL No.		
Mfr and Mfr Identification No.		
FSN		

(d) Preamplifier

	Capacitance Output units Per input unit	Resistance
1/ Transfer factor _____		
2/ Max. input load		
2/ Min. input load		
2/ Max. output load		
2/ Min. output load		
Input		
Output		
APL No.		
Mfr and Mfr Identification No.		
FSN		

(e) Preamplifier 3/

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Prior to First article inspection and quality conformance inspection, the supplier shall submit a complete testing agenda and test procedures; such as step-by-step procedure to set-up and record data, as well as data sheets, and test equipment to the Ship Control, Navigation Section of the Naval Ship Engineering Center for approval. The procedures shall consist of the detailed steps the technician will use to conduct each test.

1/ Charge to current, charge to voltage, voltage to voltage etc.

2/ Maximum and minimum values still within range response tolerance.

3/ Give same information for other type preamplifiers that may be used with the signal normalizer.

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4.2 First article inspection. First article inspection shall consist of the examination and tests specified in table I and shall be performed in general in the order listed.

Table I - First article inspection

Tests	Requirement paragraph	Test reference
General examination	3.2	MIL-I-983
Operating	3.4	4.4.2.6
Supply line voltage and frequency variation	3.2	MIL-I-983
Radio frequency noise	3.6	4.4.3
Temperature and humidity	3.2	MIL-I-983
Accelerated life	3.5	4.4.1
Dielectric strength and insulation resistance	3.2	MIL-I-983
Enclosure	3.2	MIL-I-983
Vibration (vital)	3.2	MIL-I-983
Shock	3.2	MIL-I-983

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the groups A and B inspection of table II.

4.3.1 Group A. All equipment shall be subjected to the Group A examination and tests specified in Table II. The Tests shall be performed in the order listed.

4.3.2 Group B. Whenever the basic design of the equipment, or the material of a vital part has been changed, one complete amplifier system shall be selected and subjected to the Group B tests specified in table II when required by the Command or Agency concerned (see 4.5).

Table II - Quality conformance inspection

Test	Requirement paragraph	Test reference
GROUP A		
General examination	3.2	MIL-I-983
Operating (at room environment)	3.4	4.4.2.6
Dielectric strength and insulation resistance	3.2	MIL-I-983
Supply line voltage and frequency variations	3.2	MIL-I-983
GROUP B		
Radio frequency noise	3.6	4.4.3
Temperature and humidity	3.2	MIL-I-983
Accelerated life	3.5	4.4.1
Enclosure	3.2	MIL-I-983
Vibration (vital)	3.2	MIL-I-983
Shock	3.2	MIL-I-983

4.4 Test procedures. The additional tests and test procedures specified in 4.4.1 through 4.4.2.6.3 shall be included in the data specified in 4.1.1 and shall be performed to determine conformance with the requirements of this specification. The operating tests shall apply to the preamplifier systems (see Figure 1).

4.4.1 Accelerated life test. The environmental conditions for the accelerated life test shall be cycled in accordance with Figure 2. The cycling is based on change from the high temperature plateaus to the low temperature plateaus and back to the high temperature plateaus. The temperature plateaus are numbered with the low temperature plateaus being odd numbers and high temperature plateaus having even numbers. Two hundred forty plateaus are required for the accelerated life test. Corresponding voltage, frequency, and humidity requirements are shown for a representative 10 temperature plateaus. They shall repeat in the same sequence for the entire 240 plateaus. The voltage, frequency, and humidity requirements shall occur relative to the temperature plateaus as shown on Figure 2. The system batteries shall be removed during all accelerated life testing.

4.4.1.1 High temperature plateau. The Figure 2 plateau high temperature minimums shall be 50°C for high temperature plateaus designated for testing in 4.4.1.2.1 and 65°C for high temperature plateaus not specified for testing in 4.4.1.2.1.

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4.4.1.2 Low temperature plateau. The accelerated life test low temperature plateau shown on Figure 2 shall be 10°C on the plateaus specified for testing in 4.4.1.2.1 and 2°C for low temperature plateaus not specified in 4.4.1.2.1.

4.4.1.2.1 Test frequencies. Operating test in accordance with 4.4.2 shall be required on accelerated life cycling temperature plateaus 9, 10, 11, 12/21, 22 23, 24/33, 34, 35, 36/121, 122, 123, 124/193, 194, 195, 196/201, 202, 203, 204/213, 214, 215, 216 as shown on Figure 2. Before and after the temperature cycling portions of the accelerated life test, an operating test in accordance with 4.4.2 shall be conducted under room conditions and with a 115 volt ± 1 percent, 60 c.p.s. ± 1 percent power supply. On all temperature plateaus not specified for operation testing, the system power supply shall be cycled "OFF" for 15 minutes and "ON" for 15 minutes repetitively and vibrated at 50 Hertz continuously in accordance with the type I endurance test of MIL-STD-167.

4.4.1.2.2 Voltage, frequency and humidity shall be cycled in accordance with Figure 2.

4.4.1.3 Signal input. When not testing, a 1000 Hz 1-volt signal shall be applied to the preamplifier input continuously.

4.4.1.4 All units of the amplifier system shall be sealed by the inspector before starting the accelerated life test. If the seals are opened, the accelerated life test shall start over again from the beginning. Repair or adjustment of any controls other than the operating controls specified herein constitutes failure of the accelerated life test. Three failures of accelerated life test shall require redesign subject to approval by the Ship Control, Navigation Section of the Naval Ship Engineering Center. Not more than three design changes will be approved by the Ship Control, Navigation Section, of the Naval Ship Engineering Center.

4.4.2 Operating test data.

4.4.2.1 Test accuracy. The error band of the data measured with the test instrumentation shall be less than 10 percent of the specified error or tolerance band for the unit or amplifier system under evaluation.

4.4.2.2 Chart recordings. The smallest chart division line spacing shall be greater than 1/32 inch. The smallest chart division shall be equal to one-half the error band of the data, or less.

4.4.2.3 Test density. The density shall be a fraction of rate-of-change of the characteristic being evaluated and shall double when within at least 2 percent and 5 percent of the upper and lower specified limits, respectively.

4.4.2.4 Range. Data shall be recorded at least 5 percent of the band above the specified upper limit and 50 percent of the specified lower limit below the specified lower limit or until the specified error (or tolerance) band has been exceeded by 100 percent, whichever is greater.

4.4.2.5 Scan recording. The sweep rate for a system response shall be such that the recorder deflection from zero to full scale and return to zero can take place before the scanning input has changed more than 0.1 percent of the total width of the band.

4.4.2.6 Operating test. The operating test shall demonstrate conformance with 3.4.

4.4.2.6.1 Range. Levels 10, 20, and 50 microvolts r.m.s. and levels 0.1, 0.75, 1.00, and 1.2 volts r.m.s. shall be included in the preamplifier input signal levels for recording frequency response.

4.4.2.6.2 Noise level with signal present. Unless otherwise specified (see 6.2), this test shall be conducted on the sample for first article inspection only. At 0, 100, microvolts and at 1 volt r.m.s. preamplifier input signal levels, photograph a cathode ray oscilloscope synchronized to the input test signal. Observe and record at 40, 1000, and 8000 Hz on temperature plateaus (see 4.4.1.2.1) 9, 10, 11, 12, and 213, 214, 215, 216, and on the post accelerated life room condition operating test specified in 4.4.1.2.1. The photograph shall be included in the test report at a circular size of at least 3-1/2 inches in diameter. When making these pictures, the intensity of the signal on the scope shall be increased to the point just before sharp focus is lost and photographed to clearly show the noise or "grass", and its level relative to the signal. An additional set of photographs under "room" conditions when operating on the battery power source is also required. During accelerated life temperature cycling the picture shall be taken at approximately the above frequencies during the scan and a marker shall be recorded on the frequency analysis chart paper synchronized with the camera shutter.

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4.4.2.6.3 After the accelerated life test is completed, the operating tests specified in 4.4.2.6.1 and 4.4.2.6.2 shall then be conducted using the battery power source after having been left in operation for 11 hours on battery supply to demonstrate conformance with 3.4.1.8.

4.4.3 Electromagnetic interference test. The amplifier system shall be subjected to the electromagnetic interference test of MIL-STD-462 to determine conformance with 3.6.

4.5 The results of each test shall be verified to determine conformance with this specification. In the event of failure to conform with this specificaton for any examination or test(s), the contractor shall correct promptly deficiencies discovered by him or of which he is notified.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.3).

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Basic equipment or item.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number of manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. The onboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. Where special requirements not covered by 5.1 are required, the levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made (see 6.2).

5.2.1.1 Preservation, packaging, packing and marking. The equipment and accessories, repair parts and technical publications shall be preserved and packaged levels A or C; packed level A or B as specified (see 6.2) and marked in accordance with MIL-E-17555.)

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and container (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

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5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. This general purpose audio amplifier system is for use where very low noise level and very high input impedance is required. For example, where capacitive type pick-ups (piezo-electric accelerometers, capacitance microphones and so forth) are used. Accessories are specified herein as optional items to provide for calibration of a vibration pick-up system and a microphone.

6.2 Ordering data. Procurement documents should specify the following:

(a) Sample for first article inspection (see 3.1).
(b) Type of preamplifier (see 3.3(a)).
(c) Subunits (d) and (e) of 3.3.1 when specified (see 3.4.1.10 and 3.4.1.11).
(d) Preamplifier gain (see 3.4.1.12).
(e) Preamplifier gain selection switch (see 3.4.2.1).
(f) 50-foot length of cable, when specified (see 3.4.3.1.4).
(g) Quick disconnecting type connector for signal output cable when specified (see 3.4.3.1.4).
(h) One signal output connector, when specified (see 3.4.3.2.3).
(i) Two-foot cable assembly, when specified (see 3.4.3.2.4).
(j) Other dimensions for Signal Normalizer (see 3.4.4.2.3).
(k) Noise level test when required on other than First article sample (see 4.4.2.6.2).
(l) Level of preservation, packaging, packing, and marking other than that required by 5.1 (see 5.2 and 5.3).

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 6625-N367)

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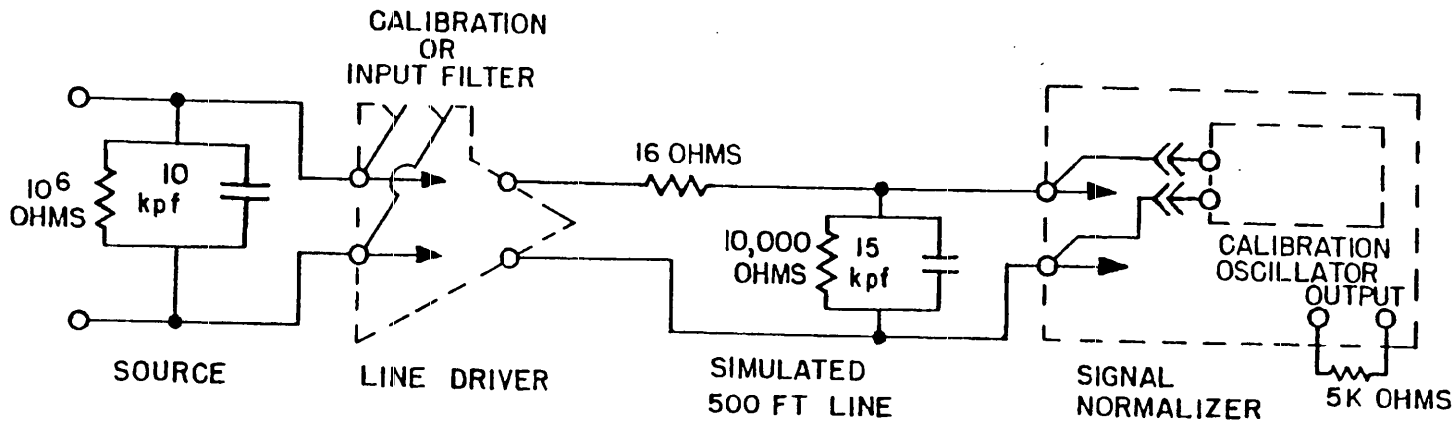


FIGURE 1 – PREAMPLIFIER SYSTEM TEST SCHEMATIC WIRING DIAGRAM

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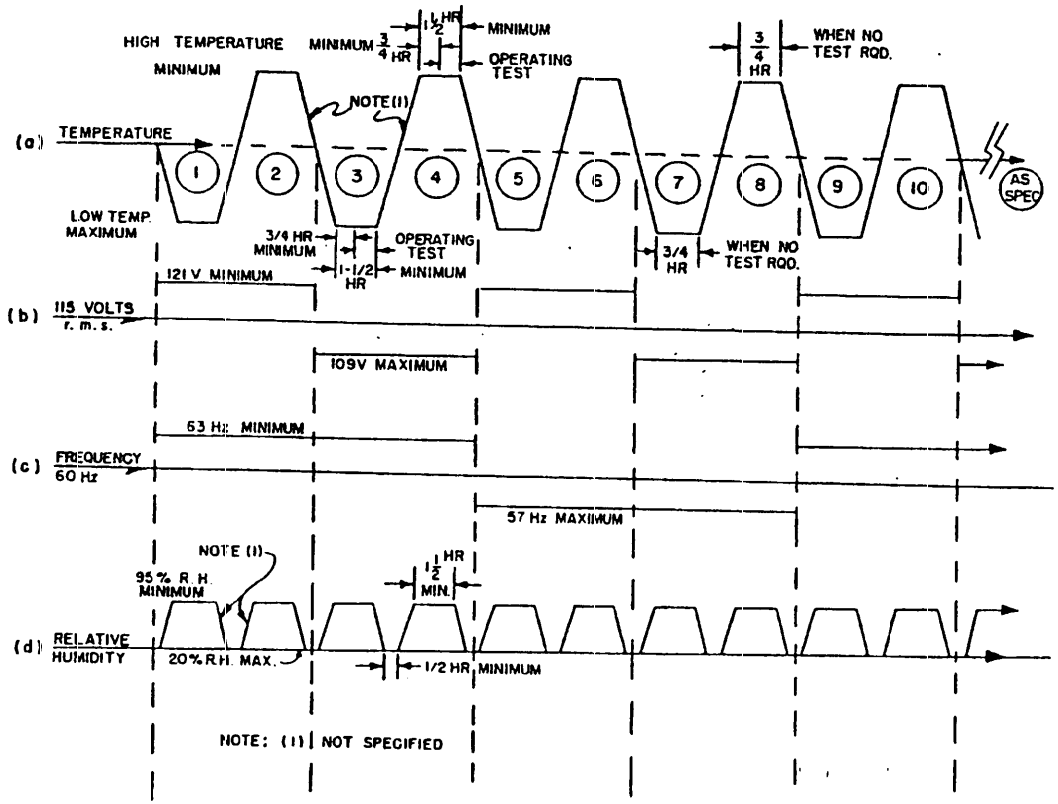


FIGURE 2- ACCELERATED LIFE TEST CYCLE DIAGRAM

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated.. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 618566
(NAVSHIPS 0VPRT 12-66)

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